MEMORANDIUM

TO: Executive Leadership Team, Whole Foods Market, Inc.

FROM: Climate & Sustainability Team, Whole Foods Market, Inc.

DATE: Friday, December 5, 2025

SUBJECT: Proposal of Carbon Credit Portfolio

I. INTRODUCTION

Whole Foods Market (WFM) serves customers with high-quality natural and organic foods from transparent sourcing. We are focused on positive climate impact, and are aware that sustainability practices are crucial in maintaining the trust of our customers and the communities we serve. As an Amazon company, Whole Foods is also part of the Climate Pledge and committed to net-zero carbon by 2040.[1] To stay on track, we will continue prioritizing emissions abatement across operations and the value chain, while starting to invest in high-integrity carbon removals as a complement.

Today, WFM's climate efforts mostly concentrate on emissions abatement. As a partner in the EPA's GreenChill program and a leader among U.S. supermarkets in hydrofluorocarbon (HFC) phaseout, WFM has equipped 24 new stores and major remodels with primarily natural-refrigerant systems as of 2024.[2] In parallel, WFM also supports nature-based climate solutions, including backing regenerative agriculture through strict regenerative label standards and supplier engagement,[3] protecting pollinators and organic ecosystems across our produce supply,[4] and co-funding a 1,000-acre "biodiversity highway" with Mad Agriculture to restore native habitat on U.S. farmland.

Building on this foundation, this memo introduces a carbon credit removal portfolio designed to complement these existing abatement efforts. In the following sections, we will outline our portfolio goals, scenarios for credit demand, and the project selection criteria.

1.1 Goal & Scenario

Whole Foods is looking to develop a carbon dioxide removal portfolio to help maintain its image and in preparation for existing and future policies legislating greenhouse gas emissions and reporting. Our customers are conscientious not just about their health, but about the world around them, and they are willing to pay a premium to ensure that they are consuming the highest quality products. As part of this, consumers must be confident that Whole Foods is a leader in the environmental space, including on climate change. Starting next year, consumers will be able to easily scrutinize Whole Foods' emissions, because California's SB 253 law requires all companies making over $1 billion in revenue to publicly report their Scope 1 and 2 greenhouse gas emissions (starting in 2026), and their Scope 3 emissions

[1] https://www.aboutamazon.com/planet/climate-pledge
[2] https://www.wholefoodsmarket.com/mission-in-action/environmental-stewardship/built-environment
[3] https://www.wholefoodsmarket.com/mission-in-action/environmental-stewardship/regenerative-agriculture
[4] https://sustainability.aboutamazon.com/natural-resources/biodiversity

(starting in 2027).[5] Additionally, California's SB 261 law requires companies to report on their climate risk and mitigation strategies, beginning January 1, 2026.[6] While there are not currently any requirements for reducing emissions or demonstrating a particular degree of climate mitigation, it seems possible that these may be developed in future years.

Developing a carbon dioxide removal portfolio that includes a range of both short- and long-term removal strategies will signal Whole Foods' proactive commitment to climate health to consumers and the government bodies administering SB 253 and SB 261. Ultimately, Whole Foods' goal should be to offset our Scope 1 and 2 emissions - which we estimate to be about 1.3 million metric tons of CO_2e* - by 2040, in alignment with the larger Amazon climate pledge. However, given the overall availability and expense of CDR credits, we believe that an achievable near-term goal is to start by reducing through insets or offsetting 20% of our Scope 1 and 2 emissions (260,000 metric tons of CO_2e) in 2026. To meet the 2040 target, Whole Foods will need to offset or reduce an additional 5-6% of emissions each year, which we think will be best be achieved through a combination of registry credit purchases, direct purchases from the projects, advance offtake agreements with newly launching projects, and investment in nascent CDR technologies through advancement market commitments made in cooperation with other companies to defray the risk to each.

Additionally, as part of the broader Amazon company, Whole Foods can leverage Amazon's existing commitments to accelerate its own carbon removal strategy. The parent company already supports early stage carbon removal through two internal mechanisms that strengthen Whole Foods' ability to secure high quality supply. The Climate Pledge Fund invests in companies developing next generation carbon dioxide removal and low carbon technologies, and several of these firms are now progressing toward commercial scale.[7] In parallel, the Sustainability Exchange platform centralizes Amazon's approach to CDR and also functions as a channel for purchasing (and offering) high integrity carbon credits, providing detailed guidance on the technologies, and its criteria for durability, additionality, and traceability of project, offering buyers a more transparent route to technologically and nature based removals.[8]

**Since Whole Foods does not publicly report its emissions at this time, our estimate for the company's Scope 1 and 2 emissions uses a competitor as a proxy. We found the reported 2021 emissions of Kroger[9] and then scaled them for Whole Foods based on each company's revenue. Whole Foods' 2023 revenue was about \$20 billion,[10] while Kroger's was about \$148 billion.[11] Accordingly, we estimated Whole Foods' emissions to be ~13.3% of Kroger's.*

[5] https://www.persefoni.com/blog/california-sb253-sb261; Whole Foods Revenue: https://www.bizjournals.com/bizwomen/news/latest-news/2024/10/whole-foods-market-amazon-growth.html
[6] https://www.persefoni.com/blog/california-sb253-sb261
[7] https://www.aboutamazon.com/planet/climate-pledge
[8] https://sustainabilityexchange.amazon.com/
[9] https://www.thekrogerco.com/wp-content/uploads/2023/07/Kroger-GHG-Goal-Roadmap_Feb-2023.pdf
[10] https://www.mashed.com/1607708/whole-foods-might-not-be-around-much-longer/
[11] https://www.macrotrends.net/stocks/charts/KR/kroger/revenue

1.2 Project Selection Criteria

Our team recommends developing a set of project criteria that reflects the company's values and our mission: to nourish people and the planet. With this in mind, we recommend project selection criteria that considers additionality, durability, leakage, social impacts, and quality. Beyond these standard key criteria, Whole Foods will also consider price as a factor, since our team must stay within our $36 million budget for GHG insets and offsets. Within these core criteria, Whole Food aims to pursue projects that have long-term storage capabilities, largely those that have a 1,000-10,000 year time horizon, to ensure that projects are durable. Additionally, we'll look to ensure that projects have very little risk of leakage, so the long durability of the projects is not lost to physical or social leakage of pushing emissions elsewhere. The impact of this is skewing our portfolio towards novel approaches, including enhanced rock weathering, biochar, bio-oil, mineralization, and capture of non-carbon super pollutants. The projects recommended below will be evaluated internally, and we will also lean on the expertise of registries, such as Puro.earth and Verra, to confirm the validity, monitoring, and verification of these projects.

Whole Foods' role as a leader in the food industry, our portfolio recommendations include an emphasis on projects that are connected to agriculture and good land stewardship. We also want to ensure that projects are located in regions connected to Whole Foods (either through a store location, or because they are part of a supply chain). We recommend projects that provide strong co-benefits for the agriculture and ranching industry, with several projects providing support for farming communities that reside below the poverty line, both in our North American store footprint and globally. Projects meeting these criteria, such as those listed below, will provide the additional benefit of serving as strategic marketing, showcasing the role that the company is playing in advancing this nascent technology while helping farmers.

II. CARBON CREDIT PORTFOLIO

2.0 Summary of Portfolio

Budget: $36 million

Insets**:** Estimated to reduce emissions by 120,000 tCO2e, at a cost of $15 million. This reduces offsets needed for 2026 to 140,000 tCO2e

Offsets (for 140,000 tCO2e):

Technology	Project (s)	# Credits to acquire	%	Price ($ per ton CO2 eq)	Total	Acquisition Pathway
Land-based Enhanced Rock Weathering	Lithos	12,000	8.6%	$ 346	$ 4,152,000	Direct purchase from project
	Terradot	12,000	8.6%	$ 346	$ 4,152,000	Direct purchase from project
	Eion	12,000	8.6%	$ 346	$ 4,152,000	Direct purchase from

						project
Biochar	Exomad Green; Aperam BioEnergia; and similar	5,000	3.6%	$ 150	$ 750,000	Purchase from Puro.Earth and Isometric
Bio oil	Charm Industrial	500	0.4%	$ 568	$ 284,000	Purchase from Isometric
DAC+ Mineralization - Geological	Cella Mineral Storage	5,600	4%	$ 650	$ 3,640,000	Frontier AMC
Mineralization-products	Neustark	1,700	1.2%	$ 200	$ 340,000	Multi year commitment or OTC from Gold Standard Registry
NbS - Reforestation and Afforestation	Mombak	11,000	7.9%	$ 50	$ 550,000	Symbiosis AMC
	Chestnut carbon	15,000	10.7%	$ 34	$ 510,000	OTC from Verra or Gold Standard Registry
NbS- Blue Carbon	Delta Blue Carbon	65,000	46.4%	$ 29	$ 1,885,000	Multiyear Commitment
Total:	140,000 credits				**$ 20,415,000**	-

Total Cost of Insets + Offsets: $35,415,000

Budget Buffer: 1.6%, to allow for cost fluctuations

2.1 Beyond Carbon: Refrigerants and HFC Phasedown

HFCs are the most widely used refrigerants in cooling and refrigeration. While having, on average, a shorter residence time in the atmosphere, they can be thousands of times more harmful to climate warming than CO_2. To cut Scope 1 emissions and reduce what needs to be covered with additional removals, WFM needs to prioritize in-value-chain cooling emissions. Specifically, we propose to aim to cut the corporate leak rate to 14% by 2030 and 10% by 2040.[12] To do so, we will (1) use natural-refrigerant systems (e.g. CO_2 or propane) for all new stores and major remodeling beginning in 2026, and (2) convert at least 35% of the current refrigerant stock to low-or-ultra-low-GWP dominant systems by 2035. Considering some sites may not be suitable for full conversion, we will do lower-GWP retrofits. Across all properties, we will implement periodic leak sensing and commit to end-of-life gas

[12] The typical food retail store refrigeration system leaks an estimated 20-25% of refrigerant annually. https://www.epa.gov/sites/default/files/2015-09/documents/gc_leaktightnesscommercialequipinstall_fact_sheet_20130912.pdf

recovery, reclaim, or verified destruction. Taken together, this pathway moves WFM well ahead of the U.S. Kigali/AIM HFC national phase-down plan and reduces our exposure to volatile HFC prices and future regulation, while strengthening our pioneering position as a climate-aware retailer.

Our team has identified a set of potential partners known for their reliability and scale. For new stores and remodels, we will source CO_2 rack systems from Carrier Commercial Refrigeration, which has deployed more than 20,000 CO_2 systems globally and offers supermarket-scale racks like MaxiCO_2OL and MiniCO_2OL.[13] At stores where CO_2 conversion is not feasible, we will retrofit legacy systems to Honeywell Solstice low-GWP HFO blends,[14] which are estimated to cut total GWP by approximately 60%,[15] while also reducing energy use up to 15%[16] and shrinking Scope 2 emissions. For end-of-life retirements, we will partner with Hudson Technologies,[17] one of the largest EPA-certified reclaimers in the U.S., who has multiple reclamation facilities and provides on-site recovery services. Recovered gas will be reclaimed under the AHRI standard, while unreclaimable gas will be properly destroyed for verified avoided emissions. WFM will continue to be an active member of the EPA's GreenChill program, benchmarking performance against industry peers.

While certified reclamation does not generate separate carbon credits, when combined with retrofits, natural refrigerant conversion, and leak-rate monitoring, our Scope 1 refrigerant emissions is estimated to be cut by approximately 120,000 tCO_2e/year by 2030 and around 350,000 tCO_2e/year by the mid-2030s.[18] That is roughly 20% of our estimated total Scope 1 and 2 footprint, directly shrinking the volume of carbon credits needed from long-duration removals. This refrigerant emission reduction pathway will require around $15 million per year investment over the next decade, depending on conversion pacing and unit costs. This corresponds to an implied abatement cost of roughly $125 per t$CO_2$e avoided. MRV are ensured through metered refrigerant flows, certified recovery and destruction, and third-party benchmarking through GreenChill.

2.2 Land-based Enhanced Rock Weathering

One innovative approach to carbon dioxide removal is enhanced rock weathering (ERW): the crushing up naturally carbon-sequesting rocks to increase the amount of surface area exposed to carbon in the air, and thus increase chemical reactions that sequester carbon.[19] By crushing up and dispersing this rock, ERW vastly increases the pace at which the rock absorbs carbon, which has proven to be far less expensive and controversial than alternative forms of CDR.[20] Beyond its carbon sequestering benefits, land-based ERW has significant co-benefits for farmers because when done on agricultural land, it improves soil health and

[13] https://www.ccr.com/offerings
[14] https://www.solstice.com/us/en/campaign/supermarkets
[15] https://www.honeywell.com/us/en/press/2021/10/honeywell-solstice-low-global-warming-potential-technology-reduces-global-greenhouse-gas-emissions
[16] https://www.gasworld.com/story/whole-foods-market-adopts-honeywell-refrigerant-technology/2091545.article
[17] https://www.hudsontech.com/refrigerant-reclaim
[18] Detailed assumptions and calculations in Appendix I
[19] Beerling, D.J., Kantzas, E.P., Lomas, M.R. et al. Transforming US agriculture for carbon removal with enhanced weathering. Nature 638, 425–434 (2025). https://doi.org/10.1038/s41586-024-08429-2
[20] https://neg8carbon.com/enhanced-rock-weathering-how-it-works/5

crop yield.[21] ERW-appropriate rocks, like basalt and olivine, remineralise soil and enhance soil fertility, decreasing the need for emissions intensive and environmentally damaging fertilizers. For marketing purposes, this is valuable because it can reduce costs and increase outputs for farmers, while helping them produce "greener" food. In addition to these co-benefits to our industry and supply chains, this solution is low cost in comparison to novel CDR methods like direct air capture (DAC), at €50–€200 per ton for EWR versus €300–€1000 for DAC.[22] This inexpensive approach also has the ability to scale to be a major source of carbon sequestration, removing up to 2 to 4 gigatonnes of CO_2 per year by 2050.[23] Based on the co-benefits, low-cost, and potential for impact, ERW could be an important component of the Whole Foods CDR portfolio.

There are a number of EWR projects for Whole Foods to select from if moving forward with this technology within the portfolio. Among the emerging leaders in scaling land-based enhanced rock weathering among farms, three companies stand out as potential offtakers to contract with: Lithos, Terradot, and Eion. These growing companies offer a range of geographic and technological approaches for how to apply ERW at scale. Lithos, a company focused on pioneering a novel measurement technique that more accurately quantifies permanent carbon removal from ERW, spreads superfine crushed basalt on farmlands.[24] Similarly, Terradot[25] applies crushed basalt to farmland in Brazil. Eion, on the other hand, provides a pelletized basalt product to be applied by farmers and ranchers to increase carbon uptake in the soil.[26] Each of these companies provides a slightly different approach to scaling ERW and improving soil health in the process. To inform our own recommended offtake agreements and ensure credibility of these credits as we refine Whole Foods' internal purchasing process, we've focused particularly on projects that the Advanced Market Commitment (AMC), Frontier Climate, has vetted.

These three ERW companies, Lithos, Terradot, and Eion, will provide a balanced portfolio for Whole Foods as it enters the enhanced weathering space. Our team recommends procuring offtake agreements for about 12,000 tons of carbon removal from each company, totaling 36,000 tons of carbon credits. This would make up about 25.8% of the overall Whole Foods CDR portfolio for permanent carbon removal. Assuming that Whole Foods would be able to negotiate with each of the three companies to achieve similar prices to those that Frontier is paying, then the total cost for Whole Foods would be $12,456,000.[27] While this is a high price tag in comparison to some nature-based climate solutions, this permanent removal would help to scale the ERW industry and could lead to millions of dollars in co-benefits for farmers and ranchers who supply Whole Foods. There is tremendous marketing potential for the company to advertise its role to develop this nascent industry, support farmers by increasing their yield and reducing their fertilizer consumption, and source products grown more "cleanly."

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[21] https://neg8carbon.com/enhanced-rock-weathering-how-it-works/5
[22] https://neg8carbon.com/enhanced-rock-weathering-how-it-works/5
[23] https://www.ipcc.ch/report/ar6/syr/downloads/report/IPCC_AR6_SYR_FullVolume.pdf
[24] https://frontierclimate.com/portfolio/lithos
[25] https://frontierclimate.com/portfolio/terradot;https://frontierclimate.com/writing/terradot
[26] https://frontierclimate.com/writing/eion;https://frontierclimate.com/portfolio/eion
[27] https://frontierclimate.com/pathway

2.3 Biochar and Bio-oil

2.3.1 Biochar

Biochar and bio-oil will both be valuable additions to Whole Foods' CDR portfolio, but they will fill different niches. Although biochar and bio-oil are both produced through the process of pyrolysis (and, in fact, are co-products), biochar is an established technology with a growing market share, while bio-oil remains an emerging technology.

Biochar projects tie to the agricultural focus of Whole Foods' portfolio on two fronts: biochar can be produced from agricultural waste products, and it can be added back to agricultural fields as a soil amendment. As a soil amendment, biochar can help soil better retain both nutrients and water, which can reduce contaminating run-off, reduce the need for synthetic fertilizers, increase drought resistance, increase soil structural stability (by feeding beneficial microbes) which can reduce erosion, and increase crop yields.[28] The Isometric carbon registry protocol states that biochar used as a soil amendment sequesters carbon in a stable form for at least 200 years.[29] Biochar can also be stored for up to 1,000 years by incorporation into building materials (like concrete or brick) or burial in a low-oxygen environment, although human meddling (such as tearing a building down, or disturbing a site where biochar was buried) could reduce this.[30] Biochar for CDR purposes is beginning to be deployed at scale, meaning biochar project credits are available now to help Whole Foods meet its CDR targets in 2026. While projects where the biochar is added to the soil are on the shorter end of the durability spectrum for CDR, it is a time horizon that is long enough to be meaningful both scientifically and to the public.

To enable the company to enter the CDR space confidently in the near term, for the next couple of years Whole Foods should buy "off-the-shelf" biochar credits that have been vetted by a CRD registry such as Puro.Earth. This can allow Whole Foods to be fairly confident of the quality; do relatively little vetting work themselves (allowing for a quick entry into the space); and shift risk related to vetting of the project to the registry, creating an extra layer of accountability in case there are any issues. Puro.Earth has accredited many projects, including Exomad Green (using sawmill waste derived biochar to repair soil in Bolivia)[31] and Aperam BioEnergia (using forest waste derived biochar to recondition soil In brazil).[32] Several other companies anticipate concluding their verification process with Isometric and beginning to issue credits by the end of 2025. These companies include: Varhad (biochar from industrial cotton stalks), Carba, and Pacific Biochar.[33]

In the immediate future, Whole Foods could purchase 5,000 -10,000 credits, at around $150 per credit.[34] This relatively small number is because projects certified by Isometric and Puro.Earth typically have less

[28] https://www.climatehubs.usda.gov/hubs/northeast/topic/digging-biochar
[29] https://registry.isometric.com/module/biochar-storage-soil-environments/1.2#durability-of-biochar-in-soils
[30] https://registry.isometric.com/module/biochar-storage-low-oxygen/1.0; https://registry.isometric.com/module/biochar-storage-built-environment/1.0
[31] https://puro.earth/CORC-co2-removal-certificate/supplier-listing/riberalta-119
[32] https://puro.earth/CORC-co2-removal-certificate/supplier-listing/aperam-bioenergia
[33] https://registry.isometric.com/suppliers
[34] https://www.spglobal.com/commodity-insights/en/news-research/latest-news/electric-power/111325-biochar-carbon-credit-prices-steady-but-sentiment-weakens

than 10,000 credits available, and sometimes less than 100.[35] Given the number of projects under development, biochar credits should become much more widely available within a few years. As this happens, Whole Foods should plan for biochar to be a gradually increasing part of its portfolio, replacing less durable and harder-to-measure nature-based solutions. To help catalyze this movement and potentially lock in a competitive price for biochar credits, Whole Foods could consider forward purchases.

2.3.2 Bio-oil

While - like biochar - bio-oil can be produced using agricultural residues, it does not have any functionality after production beyond carbon storage. Bio-oil is typically stored underground: Isometric has certified standards for storage in both salt caverns and permeable reservoirs.[36] Similarly to biochar credits, there are a limited number of bio-oil credits available right now and many fewer bio-oil projects are under development so the bio-oil market is likely to expand more slowly than the biochar market. Charm Industrial and NULIFE GreenTech both have North American projects certified by Isometric, and Corigin expects to start issuing credits in early 2026 for a project that may include bio-oil.[37] Rubicon Carbon lists the average price of bio-oil credits as $568/tonne.[38] Although bio-oil is not yet competitive on a cost front, Whole Foods will make a small, near-term investment in it, with an eye towards forging relationships that will give us priority purchasing options as the projects expand and issue more credits.

2.4 Mineralization

Considered one of the most durable removals available, WFM's parent company is already using this technology as part of its broader climate strategy, both in geological storage and in products. Through the Climate Pledge Fund, Amazon has invested in 44.01,[39] a company that injects dissolved CO_2 into peridotite and other ultramafic rocks to mineralize the carbon as a stable carbonate, providing very long durability periods. On the product side, Amazon has also invested in CarbonCure,[40] which injects captured CO_2 into fresh concrete where it reacts with calcium and forms mineral carbonates that remain stored long term. These existing relationships mean that mineralization is already recognized by the company as a high-quality credit. The portfolio section below therefore focuses on complementary partners that are not currently tied to Amazon, so that WFM can add incremental, clearly attributable removals rather than simply scaling projects that the parent company already anchors.

2.4.1 Product-based Mineralization

Neustark offers a complementary pathway that does not overlap with Amazon's existing CarbonCure investment. Neustark delivers biogenic CO_2 to demolition concrete recyclers, where the gas mineralizes into stable carbonates within the aggregate[41]. These credits are certified under the Gold Standard, and recent multiyear offtakes indicate prices around $200 per ton CO_2e.[42] For this portfolio, the credits could

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[35] https://puro.earth/CORC-co2-removal-certificate/supplier-listings
[36] https://registry.isometric.com/protocols?pathway=biomass-carbon-removal-and-storage
[37] https://registry.isometric.com/suppliers
[38]https://rubiconcarbon.com/blog/carbon-removal-insights-biomass-with-carbon-removal-and-storage-bicrs/
[39] https://www.4401.earth/blog/44-01-completes-usd37m-series-a-investment-round
[40] https://carboncredits.com/carboncure-carbon-removal-technologies-for-conrete-raises-over-8m/
[41] https://www.neustark.com/en/our-solution-mineralization
[42] https://www.senken.io/blog/12-leading-carbon-credit-providers-in-2025-a-selection-guide-for-companies

be purchased directly through the registry or routed through Amazon's Sustainability Exchange. Because this pathway is linked to construction and refurbishment activity, we will begin with a conservative purchase of 1,700 tons, with volumes adjusted in subsequent years as project availability or market conditions evolve.

2.4.2 Mineralization with geological storage

Cella Mineral Storage[43] injects purified CO_2 from DAC and other high quality streams into basalt formations, where it mineralizes into solid carbonate with +1000 years of stable permanence. Early projects in Kenya have already been contracted through advance market commitments from Frontier buyers, which establishes credible MRV standards and demonstrates strong technical validation. Although Cella's own pricing is undisclosed, comparable durable DAC+ mineralization contracts range from roughly \$500 to \$1000 dollars per ton[44], so this pathway should be treated as a high permanence, low volume component of the portfolio. WFM can access Cella removals either through Frontier advance market commitment or through a direct offtake added to Amazon's Sustainability Exchange. Both options allow Whole Foods to secure future tons, signal long term demand, and keep credit delivery within Amazon's internal accounting system.

2.4.3 Catalyzing the Market with Frontier

Since projects involving in-situ mineralization rely heavily on DAC, which is not yet a very cost effective solution, Whole Foods could also consider joining Frontier Climate as an avenue for purchasing DAC + mineralization credits. Participation in Frontier will allow WFM to help catalyze the DAC market with moderate investment (at least \$10 million before 2030) and minimal effort to vet projects.[45] On average, DAC credits purchased by members of Frontier cost \$650/tonne, so if Whole Foods were to make the minimum investment by spending \$3.6 million per year through Frontier, it could expect to receive about 5,600 credits per year.[46]

2.5 Nature based Solutions

Nature-based solutions (NbS) will complement Whole Foods' engineered CDR portfolio by delivering near-to-medium-term carbon removal alongside co-benefits for soil health, biodiversity, and farming communities. Given our identity as a food-first retailer and our strong ties to growers and ranchers, we propose a focused set of high-integrity NbS pathways that are rooted in agriculture and land stewardship, and meanwhile can meet the additionality and quality criteria. We see NbS as a near-term solution in the portfolio, while other novel CDRs are not yet at scale. However, we will incrementally decrease its share over time, so that NbS do not crowd out long-durable removals like ERW, mineralization and bio-oil. Aligned with Amazon's internal vetting framework, all the NbS credits we propose meet ABACUS label's rigorous standards for additionality, permanence and leakage prevention.[47]

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[43] https://www.cellamineralstorage.com/
[44] https://www.forbes.com/sites/phildeluna/2024/11/29/will-direct-air-capture-ever-cost-less-than-100-per-ton-of-co/
[45] https://frontierclimate.com/participate
[46] https://frontierclimate.com/pathway
[47] https://verra.org/program-notice/verra-launches-abacus-label-for-verified-carbon-units/

2.5.1 Reforestation : Mombak-Symbiosis (10% of NbS)

This project is Mombak's large-scale reforestation of degraded pasturelands in the Brazilian Amazon. It is the first project to pass the rigorous selection of the AMC Symbiosis Coalition,[48] and aims to restore biodiverse forests (over 80+ native species). The project is monitored with high-resolution biomass measurements and uses dynamic baselines to quantify real carbon gains. It also has a strong strategy for a permanence/durability of 100 years through degraded land purchases and rural partnership agreements.[49] Overall, the project offers multiple co-benefits for the people and the ecosystem, making it a strong fit for WFM's portfolio goal. Its credits are available for over-the-counter (OTC) purchase from Symbiosis Coalition at "over US$ 50 per tonne CO_2e."[50] However, due to Google's recent purchase for 200,000 tons of removal credits, the volume available for OTC is uncertain. Considering that, WFM should acquire any large amount of credits through forward-purchase, contracting long-term offtake agreements through Amazon Sustainability Exchange.[51] Due to its supply risks, we recommend to keep the volume of purchase at around 15% of the NbS portfolio,

2.5.2 Afforestation: Chestnut Carbon (20% of NbS)

Chestnut Carbon is a U.S.-based company focused on high quality carbon credits through afforestation and reforestation projects on historically cleared or degraded lands. For this portfolio, we will highlight the Chestnut Sustainable Restoration Megaton Project,[52] implemented and verified following the Gold Standard Land Use and Forest Activities Requirements and A/R (afforestation and reforestation) methodologies[53]. The project collaborates with landowners to reintroduce trees into mixed landscapes to increase carbon stocks, sequester greenhouse gases and enhance ecosystem services co-benefits such as soil stability and wildlife habitat protection, aligning closely with WFM's mission.

Recently, Chestnut Carbon sold 66,000 tons of credits at US$34 per tCO_2e to Microsoft,[54] making it a reliable choice. To purchase credits from this project, WFM can either directly buy from the developer or via Amazon's Sustainability Exchange, which can leverage Amazon's scale to secure a more stable offtake pricing over multiple years. Given its availability and high-integrity, we recommend that it take 40% of the NbS slice.

2.5.3 Blue Carbon: Delta Blue Carbon **(70% of NbS)**

Delta Blue Carbon (DBC) is one of the world's largest leading blue carbon projects, aiming to restore mangrove systems in regions with historical land use pressure and ecological degradation. This project aims to restore 300,000 hectares of mangrove on the south east coast of Pakistan. Mangroves sequester carbon at rates higher than most terrestrial forest, because carbon is stored not only in above ground biomass but also locked into deep water-logged soil with a long term stability. We are interested in this project, as it can provide a compelling story around coastal stewardship and sustainable seafood. Similar to the other two projects, credits for DBC can be purchased via Amazon Sustainability Exchange with a

[48] https://www.symbiosiscoalition.org/perspectives/catalyzing-high-integrity-nature-based-carbon-removal
[49] https://registry.verra.org/app/projectDetail/VCS/5389
[50] https://mombak.com/news/carbon-credit-maker-mombak-signs-deal-with-mclaren-good-pricing-news/
[51] https://sustainabilityexchange.amazon.com/
[52] https://www.carbon-intel.com/projects/gold-standard/23278/chestnut-sustainable-restoration-megaton-project-united-states
[53] https://registry.goldstandard.org/projects/details/5090
[54] https://landreport.com/chestnut-carbon-microsoft-offtake-agreement

price close to the auction prices of around \$29 per tCO_2e, making it the most cost effective removal credits in our portfolio and thus is recommended to take up 45% of the NbS portfolio.

2.6. Risks and Mitigation Factors

Uncertainty of Regulatory and Market Standards

As the current standards for carbon removal are still being developed, the credits proposed in this portfolio could risk not meeting future regulatory expectations. Nature Based projects are particularly exposed, as their requirements for permanence, leakage and baseline guidelines are already being highly scrutinized.[55] To mitigate this risk, this proposal is aligned with ABACUS label's strong integrity standards, as well as either anchored to the highest integrity registries such as Isometric, Puro.Earth and Gold Standard, or vetted through Frontier and Symbiosis, which face continuous methodological review.

Delivery and Scaling constrains

Engineered CDR pathways such as DAC+ mineralization and bio-oil, face considerable risks from scaling delays, and supply chain limitations that could limit delivery volumes and timelines. Mineralization into product projects is highly dependent on the expansion and maintenance activities of WFM, and ERW projects face seasonal variability constraints that further increase the uncertainties in measuring its carbon uptake. To address this risk, the distribution of this portfolio includes near-term available credits from biochar, and NbS projects as well as forward staged commitments tied to verified delivery.

Durability and Reversal Risks

NbS projects have short durability periods and are highly vulnerable to shifts in land management, fire, drought and policy decisions that affect land tenure. On the other hand, engineered and geochemical pathways offer longer durability periods, they rely on strict technical conditions: ERW durability is long but depends on accurate quantification of soil processes; biochar used as soil amendment is very sensitive to soil chemistry disturbance; and bio-oil and geologic storage methods require very strict monitoring. To mitigate these risks, this portfolio included projects with legally enforceable land-use agreements, long-term monitoring, and dynamic baselines, and balanced the selection of projects, alternating NbS and projects with high-durability engineered removals.

Measurement, Verification, and Transparency Limitations

Accurate measurement and transparent reporting are essential to reduce credit quality risks, since each pathway introduces its own sources of uncertainty. ERW requires precise quantification of carbon dissolution, and results vary across developers depending on how soil reactions are measured, while biochar and mineralization pathways introduce different MRV challenges. Biochar MRV changes with feedstock type and application method, and mineralization projects rely on subsurface geochemical modeling that must be validated against long term monitoring data. To address these risks, this portfolio incorporates the internal audit consistency provided through Amazon's Sustainability Exchange and prioritizes purchases from projects with strong third party verified protocols and established MRV, such as

[55] Buma, B., Gordon, D.R., Kleisner, K.M. et al. Expert review of the science underlying nature-based climate solutions. Nat. Clim. Chang. 14, 402–406 (2024). https://doi.org/10.1038/s41558-024-01960-0

Isometric for biochar and bio oil, Gold Standard for Neustark, and Frontier reviewed methodologies for Cella.

Portfolio-level Risk and Distribution

At the portfolio level, WFM must balance credit availability, durability and exposure to cost volatility across multiple technologies. Engineered removals such as ERW, mineralization, and bio oil offer high durability but remain capacity constrained and expensive, which creates risks to supply shortages, delivery delays, and increasing marginal costs. By contrast, nature based solutions provide near term volume at lower cost but introduce durability and reversal risks that limit their long term climate value if used too heavily.

The present portfolio mitigates these risks by layering technologies with different maturity levels, geographies, feedstocks, and permanence profiles. We combine near term, registry verified supply from biochar and early ERW delivery with longer duration pathways such as mineralization, bio oil, and DAC mineralization secured through controlled forward offtakes and pooled mechanisms like Frontier, which also reduces non delivery or price escalation risks. We also source credits across different regions (North America, Latin America, Europe, and South Asia), ensuring that disruptions in any one region or feedstock do not materially compromise delivery or expenditures.

Conclusion

This portfolio secures a mix of diversified, high-integrity removal for WFM in 2026, encompassing projects from EWR, biochar and bio-oil, mineralization, and NbS, blending near-term purchases with multiyear offtakes. It totals to 140,000 tCO_2e removals with a budget of $36 million. In parallel, we expect to deliver around 120,000 tCO_2e/year insets, primarily from refrigerant management and HFCs phasedown, at an estimated cost of $15 million per year. Projects are chosen to minimize risk, provide social and ecological safeguards, and emphasize additionality and permanence, all strongly resonating with WFM's core values in food first identity and climate actions.

Taken together, our proposed measures effectively reduce and offset 20% of WFM's predicted annual corporate CO_2e emissions with total costs of around $35.4 million, with a small budgetary buffer for risk management purposes. It is a meaningful step towards meeting WFM's net-zero by 2040 under the Climate Pledge. As the CDR market matures, we will progressively increase the shares of higher-durability options like EWR, bio-oil, and mineralization, while keeping within-value-chain insets as our priority. Above all, WFM is committed to keep this portfolio transparent, robust, and worthy of customer trust.

Group Contributions

Group members brainstormed the scenario and goal together, and then wrote the following sections:

- Jhoana: Mineralization; Risks and Mitigation Factors; section of Nature-based solutions
- Rose: Goal & Scenario; Biochar & Bio-oil; Frontier (section of Mineralization); slide/paper edits
- Ruiyan: Introduction; Nature-based Solutions; HFCs; Conclusion
- Taylor: Project Selection Criteria; Enhanced Rock Weathering; draft of slides

Appendix I. Refrigerant Emissions and Abatement Cost Calculation

A. Baseline

We assume:

- WFM has 530 stores in the U.S[56].
- Average store floor size is 40,000 square feet[57]
- Average refrigerant stock per store is 1,400 kg (estimates based on floor area)
- Total installed refrigerant volume: Q_{total} = 750,000 kg[58]
- Baseline annual leak rate: 20% of installed stock
- Legacy refrigerants have average GWP around 2,500

Baseline annual refrigerant emissions from leakage:

$$E_{baseline} = Q_{total} * 0.20 * GWP_{legacy}/1000 \approx 375,000\ tCO2e/year$$

Which is roughly 30% of Whole Foods' estimated 1.3 $MtCO_2e$ Scope 1+2 emissions.

B. Emissions Pathway Calculations

1. 2030: leak-rate reduced to 14%

Total annual emissions:

$$E_{2030} = 500,000 * 0.14 * 3,900/1,000 = 270,000\ \text{tCO2e/year}$$

Avoided vs baseline:

$$\Delta E_{2030} = 390,00 - 270,000 \approx 120,000\ tCO2e/year.$$

2. 2035 – 12% leak rate + 35% naturals + 63% low-GWP retrofits

Knowing that GWP is close to 0 for natural refrigerants, and low-GWP blends typically have GWP around 1400:

- Natural refrigerants stock: Q_{nat} = 0.35* 500,000 = 175,000 kg
- Low-GWP refrigerant stock: Q_{low} = 0.65* 500,000 = 325,000 kg

Total annual emissions:

$$E_{2035} = (Q_{nat} * 0 + Q_{low} * 1,400/1,000) * 0.12 \approx 55,000\ tCO2e/year$$

Avoided vs baseline:

$$\Delta E_{2035} = 390,000 - 55,000 \approx 340,000\ tCO2e/year$$

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[56] https://media.wholefoodsmarket.com/about/
[57] https://media.wholefoodsmarket.com/whole-foods-market-to-open-smaller-format-stores-as-part-of-ongoing-expansion/
[58] ChatGPT estimates

C. Cost Assumptions

Assuming[59]:

- Natural-refrigerant (CO_2) system costs $300,000–$500,000 per store (take $400,000)
- Lower-GWP retrofit: $30,000–$80,000 per site (take $50,000)
- Leak detection and monitoring: $2 million/year.

Illustrative mid-range scale:

- Conversions of 175 stores to naturals over 10 years
- 10 new stores or major remodels per year.
- 30 lower-GWP retrofits per year.

In this scenario, the total cost over 10 years can be estimated at

$$175 * 400k + 100 * 300k + 300 * 50k + 3000k * 10 \approx \$145,000,000$$

Giving annual cost around $15 million.

[59] Price estimations provided by GPTs.